

**A Controlled Back Hallway for the Four-Dimensional Whirligig:
Exact Clifford-Torus Forcing, an Elastic-Foundation Equivalence,
and an $SU(2)$ Lift**

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Abstract

The Scalar–Angular–Torsion (SAT) archive proposes a four-dimensional Whirligig action and a Universal Indicatrix (UI) trajectory generator. Rather than assuming that a generic mapping between general relativity and quantum mechanics follows from a common embedding, we isolate a testable sector. We prove an exact inverse-forcing result for two-frequency Clifford-torus trajectories, derive a nonnegative curvature defect distinguishing great circles from genuine two-frequency motion, lift those trajectories explicitly to $SU(2) \cong S^3$, and exhibit a precise Fourier-domain equivalence to a periodically forced Euler–Bernoulli beam on a Winkler foundation. A dimensionless $(m, n) = (2, 3)$ test reproduces its fourth-order Euler–Lagrange equation with a maximum numerical residual 2.5×10^{-15} , and a held-out harmonic is predicted from the other using one stiffness ratio. Linearization about the exact orbit also generates a parity-protected interplane sideband selection rule absent from the uncoupled beam. The full unforced Whirligig action does not automatically make the chosen torus paths stationary. The result is a narrow, analytically verified translation map, not a GR–QM isomorphism, a claim of mathematical priority, or evidence for a new physical interaction.

Keywords: four-dimensional curves, Clifford torus, fourth-order variational equations, inverse design, elastic foundation, $SO(4)$, $SU(2)$, spectral filtering

I. MOTIVATION AND SOURCE DISCIPLINE

The historical Whirligig/Topological Derivation Manifold and UI appear in the SAT archive as tools for translating between representations of physical systems [1, 2]. The recent user-supplied equation sheet specifies the trajectory generator $y^\mu(s) = r(s)R^\mu{}_\nu(s)x_0^\nu$ with $R \in SO(4)$ and the quadratic-bending Whirligig Lagrangian. We treat those formulas as the source model. No numerical constants from SAT’s particle-sector claims are inserted. Distinguish an exact mapping of *mathematical models* from claims that different physical phenomena are identical.

For a 2π -periodic $H : S^1 \rightarrow \mathbb{R}^4$ and prescribed $G : S^1 \rightarrow \mathbb{R}^4$, let

$$\mathcal{J}[H; G] = \int_0^{2\pi} \left[\frac{\kappa}{2} \|H''\|^2 + \frac{\lambda_s}{2} (\|H\|^2 - R^2)^2 + \frac{k}{2} \|H - G\|^2 \right] ds, \quad \kappa, k > 0, \lambda_s \geq 0. \quad (1)$$

With G held fixed during variation and periodic boundary conditions, the Euler–Lagrange

equation is

$$\kappa H^{(4)} + 2\lambda_s(\|H\|^2 - R^2)H + k(H - G) = 0. \quad (2)$$

The parameter s here is dimensionless. Physical units are supplied only when identifying a particular mechanical counterpart.

II. AN EXACTLY SOLVABLE UI SECTOR

Choose integers m, n , nonzero radii $a, b > 0$ with $a^2 + b^2 = R^2$, and define

$$H_{m,n}(s) = (a \cos ms, a \sin ms, b \cos ns, b \sin ns)^T. \quad (3)$$

This is a UI orbit with $r(s) = R$ and $x_0 = (a, 0, b, 0)^T/R$. Its rotation is $R(s) = \text{diag}[\text{Rot}(ms), \text{Rot}(ns)] \in SO(4)$. The special case $a = b = R/\sqrt{2}$ lies on the equal-radius Clifford torus in S_R^3 ; arbitrary positive a, b define a product torus within the same hypersphere. This construction uses two of the six independent generators available to $\mathfrak{so}(4)$; it makes no assumption about a unique inverse UI control history.

The constant speed and extrinsic squared curvature numerator are

$$\|H'\|^2 = a^2 m^2 + b^2 n^2 \equiv v^2, \quad \|H''\|^2 = a^2 m^4 + b^2 n^4. \quad (4)$$

Proposition 1 (A curvature-defect identity). *For the family (3),*

$$\boxed{R^2 \|H''\|^2 - \|H'\|^4 = a^2 b^2 (m^2 - n^2)^2 \geq 0.} \quad (5)$$

The defect vanishes for $|m| = |n|$ (or if one circle collapses). Thus with $a, b > 0$ and nonzero speed the constant-speed curve has zero intrinsic geodesic curvature as a curve of S_R^3 precisely when $|m| = |n|$.

Proof. Substitute the two squared norms, expand using $R^2 = a^2 + b^2$, and cancel the diagonal terms. For a constant-speed curve on a round sphere, the tangential acceleration is $H'' + (v^2/R^2)H$ and its squared norm equals $\|H''\|^2 - v^4/R^2$. Hence (5) is R^2 times the squared tangential acceleration. $\square$

Proposition 2 (Exact inverse-forcing theorem). *Define the target configuration*

$$\boxed{G_*(s) = H_{m,n}(s) + \frac{\kappa}{k} H_{m,n}^{(4)}(s).} \quad (6)$$

Then $H_{m,n}$ is an exact stationary solution of (1). If $\lambda_s \geq 0$ and $k, \kappa > 0$, its second variation is strictly positive for all nonzero periodic perturbations η :

$$\delta^2 \mathcal{J}[H_{m,n}](\eta, \eta) = \int_0^{2\pi} [\kappa \|\eta''\|^2 + k \|\eta\|^2 + 4\lambda_s (H_{m,n} \cdot \eta)^2] ds > 0. \quad (7)$$

In particular it is a strict local minimizer in the natural periodic H^2 topology.

Proof. On the chosen torus orbit $\|H\|^2 = R^2$, so the sphere-penalty gradient in (2) vanishes. Substitute (6); the remaining terms cancel pointwise. Differentiating the action twice yields (7); its $k\|\eta\|^2$ term proves strict positivity. $\square$

The target is constructive, not a postdiction: G_* is deliberately designed for a specified trajectory. Its harmonic components are amplified by $1 + (\kappa/k)m^4$ and $1 + (\kappa/k)n^4$. This means any claimed physical prediction requires *independent* information about how G is realized.

Corollary 1 (Unforced torus-trajectory obstruction). *For a nonconstant curve of the form (3) with $\|H\| = R$, setting $G = H$ in the source equation (2) leaves $\kappa H^{(4)} = 0$, which is impossible for nonzero windings. Thus the archived bending-plus-penalty action does not by itself establish that a generic two-frequency hyperhelix is a free stationary solution.*

This narrow obstruction concerns the specific model and assumptions above, not every higher-order SAT action.

III. EXACT BACK HALLWAY TO A STANDARD ELASTIC SYSTEM

An Euler–Bernoulli beam with flexural rigidity EI , resting on a linear Winkler foundation of stiffness per unit length K , has the stationary energy

$$\mathcal{E}[w] = \int_0^L \left[\frac{EI}{2} \|w_{xx}\|^2 + \frac{K}{2} \|w - g\|^2 \right] dx, \quad EI w_{xxxx} + K(w - g) = 0. \quad (8)$$

Here w may denote one transverse component or a vector of independently responding components. With $s = 2\pi x/L$, the Fourier mode e^{ins} obeys

$$\boxed{\frac{w_n}{g_n} = \frac{1}{1 + \alpha n^4}, \quad \alpha = \frac{EI}{K} \left(\frac{2\pi}{L} \right)^4.} \quad (9)$$

For the shell-respecting exact solution (6), each component of the Whirligig satisfies the identical spectral transfer with $\alpha = \kappa/k$. The resulting map between Fourier coefficients of

the two periodically forced fourth-order operators is exact and invertible on periodic Sobolev spaces with the appropriate regularity. For arbitrary variations away from $\|H\| = R$, the nonlinear penalty in (1) prevents treating the *entire* Whirligig functional as a linear beam. This is a restricted operator equivalence, not a theory-wide isomorphism.

A parameter-elimination test follows immediately. For two independently observed nonzero harmonics m, n , with known source/response amplitudes, define $A_j = g_j/w_j$. Then

$$\boxed{\frac{A_m - 1}{m^4} = \frac{A_n - 1}{n^4} = \alpha.} \quad (10)$$

A measurement of the first mode predicts the second with *no additional fit parameter*. If (10) fails outside experimental errors, this particular linear back hallway is rejected. Measurements must establish that the forcing is prescribed rather than inferred from the response.

IV. WHICH LIE GROUP ACTUALLY APPEARS?

Write $(z_1, z_2) = (ae^{ims}, be^{ins})/R$. The explicit map

$$U(H) = \begin{pmatrix} z_1 & -\bar{z}_2 \\ z_2 & \bar{z}_1 \end{pmatrix} \quad (11)$$

is in $SU(2)$: $U^\dagger U = I$ and $\det U = |z_1|^2 + |z_2|^2 = 1$. It realizes the standard round-three-sphere/unit-quaternion identification $S^3 \simeq SU(2)$. Moreover,

$$\|H''\|^2 = \frac{R^2}{2} \text{tr}(U''^\dagger U''). \quad (12)$$

Thus the embedded bending energy becomes exactly $\kappa R^2/4 \int \text{tr}(U''^\dagger U'') ds$.

The generating rotations belong to $SO(4)$, with spin cover $\text{Spin}(4) \simeq SU(2)_L \times SU(2)_R$, not $SU(4)$. This resolves the *group typing for this particular UI construction*; it does not settle the group of a separate, historical Schwarzschild-to-gauge transformation.

V. NUMERICAL CHECKS AND FALSIFICATION PATH

Set $(a, b) = (1, 1)/\sqrt{2}$, $(m, n) = (2, 3)$, $R = 1$, $\kappa = 0.02$, $k = 1$, $\lambda_s = 5$, and use 4096 equally spaced periodic sample points. The amplitude factors are

$$A_2 = 1 + 0.02(2^4) = 1.32, \quad A_3 = 1 + 0.02(3^4) = 2.62. \quad (13)$$

The geometric defect (5) is 6.25, the action is 7.3305922979 in the dimensionless normalization, and the maximum directly evaluated residual of (2) is 2.48×10^{-15} . A separate centered fourth-difference approximation converges at roughly a factor of four under doubling resolution: errors 0.20747, 0.05193, 0.01299, 0.00325 for $N = 128, 256, 512, 1024$. Nonzero perturbations increase the computed action, consistent with (7). The $SU(2)$ embedding satisfies unitarity and determinant one to $\lesssim 7 \times 10^{-16}$.

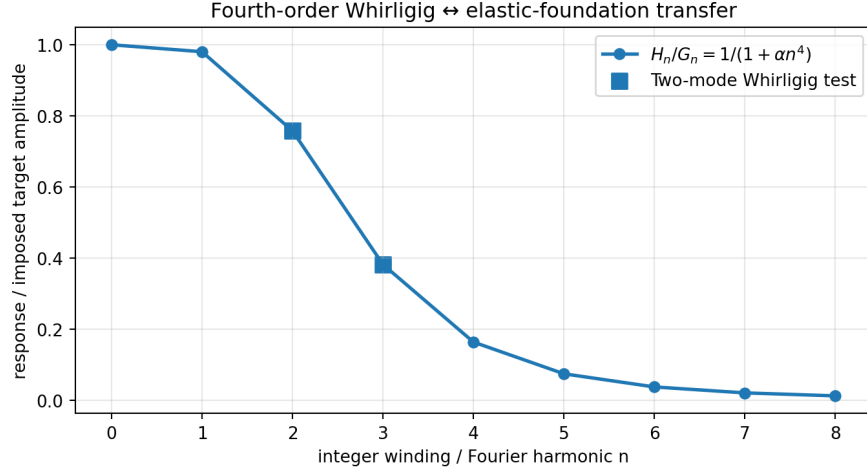


FIG. 1. Exact linear Fourier transfer for the two winding modes. The plotted response is identical to that of a periodically forced Euler–Bernoulli beam on a Winkler foundation with the indicated dimensionless stiffness ratio. These are synthetic analytic-model values, not measurements.

VI. BEYOND THE BEAM: A MODE-CONVERSION SELECTION THEOREM

The exact elastic-foundation equivalence applies along the selected on-shell orbit. The hypersphere penalty nevertheless leaves an operationally distinguishable signature upon small perturbation. Hold the designed background target G_* fixed, perturb it by $\delta G = f$, and write $H = H_0 + h$. Linearization around the exact solution $H_0 = H_{m,n}$ yields

$$\left[\kappa D_s^4 + k + 4\lambda_s H_0(s) H_0(s)^T \right] h(s) = k f(s). \quad (14)$$

The term $H_0 H_0^T$ has rank one at each s , but rotates with the background curve. It creates genuine cross-plane coupling that the diagonal linear beam operator lacks. The operator is positive on periodic H^2 perturbations for $\kappa, k > 0$, $\lambda_s \geq 0$, so the response is unique.

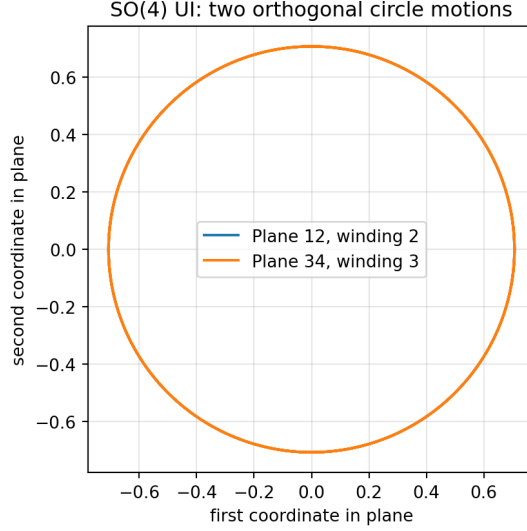


FIG. 2. The two circular coordinate motions of the four-dimensional UI generator. This plot is a two-panel-coordinate representation, not a perspective rendering of S^3 .

Proposition 3 (Half-turn harmonic selection). *Assume m even and n odd. Set $P = \text{diag}(1, 1, -1, -1)$ and $f(s) = \cos(s)e_1$. The unique solution of Eq. (14) contains only odd Fourier harmonics in the first two coordinates and only even ones in the last two.*

Proof. The background satisfies $H_0(s + \pi) = PH_0(s)$. Define the involution $Th(s) = Ph(s + \pi)$. The operator in Eq. (14) commutes with T , whereas $Tf = -f$. By uniqueness of the solution, $Th = -h$. In coordinates, $h_{1,2}(s + \pi) = -h_{1,2}(s)$ while $h_{3,4}(s + \pi) = h_{3,4}(s)$. The claimed Fourier parities follow. $\square$

For the same $m = 2, n = 3, \kappa = .02, k = 1, \lambda_s = 5$ example, we solved Eq. (14) on a 128-point periodic grid with a Fourier fourth-derivative operator and conjugate-gradient iteration, preconditioned by the uncoupled beam inverse. The relative equation residual was 1.24×10^{-11} . An independently imposed unit-amplitude $\cos s$ in the first coordinate generated amplitudes 0.57713 (winding 1) and 0.07535, 0.01585 (windings 3 and 5), while the third coordinate acquired 0.21391 (mean), 0.14956 (winding 2), 0.03226 (winding 4), and 0.00795 (winding 6). In the corresponding uncoupled beam the third coordinate remains identically zero. A separate direct dense solve using a 64-point Fourier differentiation matrix independently recovered the cross-plane winding-2 amplitude 0.1495585124046 and first-plane winding-3 amplitude 0.0753500902191, within 10^{-8} of the 128-point iterative solution; its relative equation residual was 8.81×10^{-13} . This parity-resolved interplane signal

is a *mathematically distinct* output of the nonlinear shell coupling, conditional on a real system implementing that coupling. An experimentally meaningful test must first realize and independently characterize the four-channel constraint.

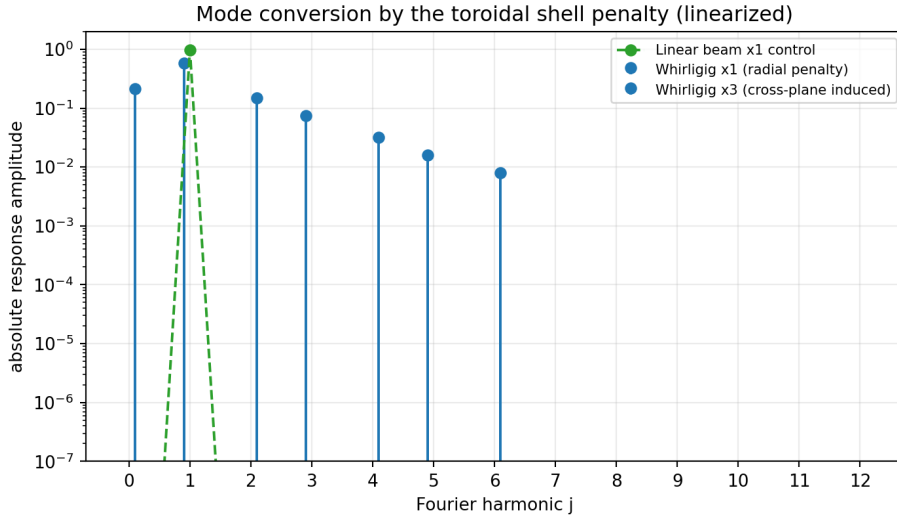


FIG. 3. Linearized Whirligig sideband conversion from one externally imposed Fourier mode in the first coordinate, compared with the same linear beam operator with the geometric sphere penalty switched off. Synthetically calculated amplitudes, not empirical observations. The predicted odd/even selection is exact for this parity class.

VII. DISCUSSION, LIMITATIONS, AND PRIORITY

The result offers a modest but genuinely closed SAT “back hallway”: a two-frequency UI orbit can be translated to a standard measurable response law without invoking black-hole metrics, hydrogen energies, unverified particle constants, or empirical fitting of two separate sectors. It identifies a mismatch in the common shorthand “the harmonic torus automatically solves the bending action,” supplies the missing forcing exactly, and isolates a curvature defect that can track whether a pair of modes represents a great-circle orbit or genuine two-frequency winding.

The mathematics in this paper inherits standard results on $S^3 \cong SU(2)$, $SO(4)$, and linear beam theory. We assert *no priority* for those identities. A historical novelty check is still required for this particular inverse-design formulation and invariant packaging. The present

sector does *not* derive Einstein’s equations, a hydrogenic spectrum, spin, quantum probabilities, a Navier–Stokes regularity theorem, or an experimental prediction unique to SAT. Especially, the spectral identity here is a beam-transfer statement, not the broad curvature–Laplace–Beltrami “hinge identity” appearing in some archival summaries. A suitable first experiment is a periodic, multi-harmonic elastic displacement system with independently commanded drive g and measured response w : infer EI/K from mode m and predict mode n with a fixed, predeclared uncertainty budget. Only afterward should one test nonlinear, coupled, or geometric departures.

DATA AND REVIEW STATUS

Complete executable Python and numerical outputs accompany this research draft. This manuscript is **SANDBOXED**; it has not undergone independent hostile peer review, external novelty verification, or experimental validation. Comparisons to externally published papers are supplied in separate review packets; they are *standards-of-evidence comparisons*, not claims of equivalent scope.

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- [1] SAT archive, *4DHH Lagrangian* and the associated Whirligig/Universal Indicatrix equation sheets, https://github.com/Satobloc/SAT_THEORY_ARCHIVE_2023-25 (historical primary source; equations independently recomputed here).
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